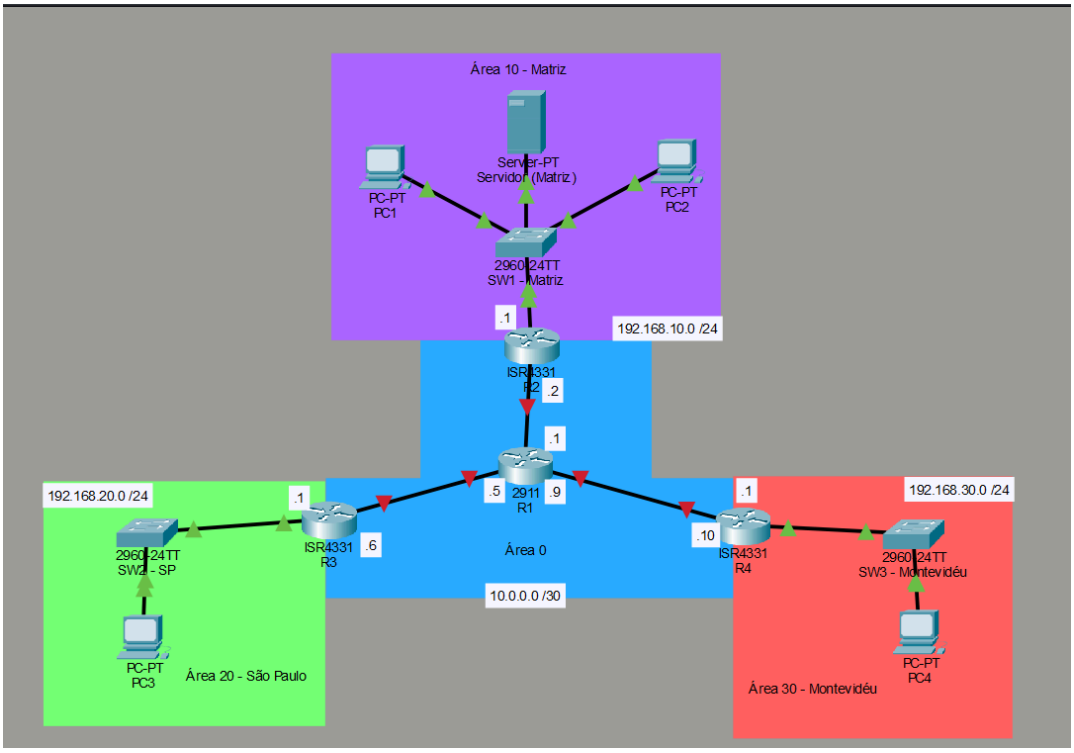
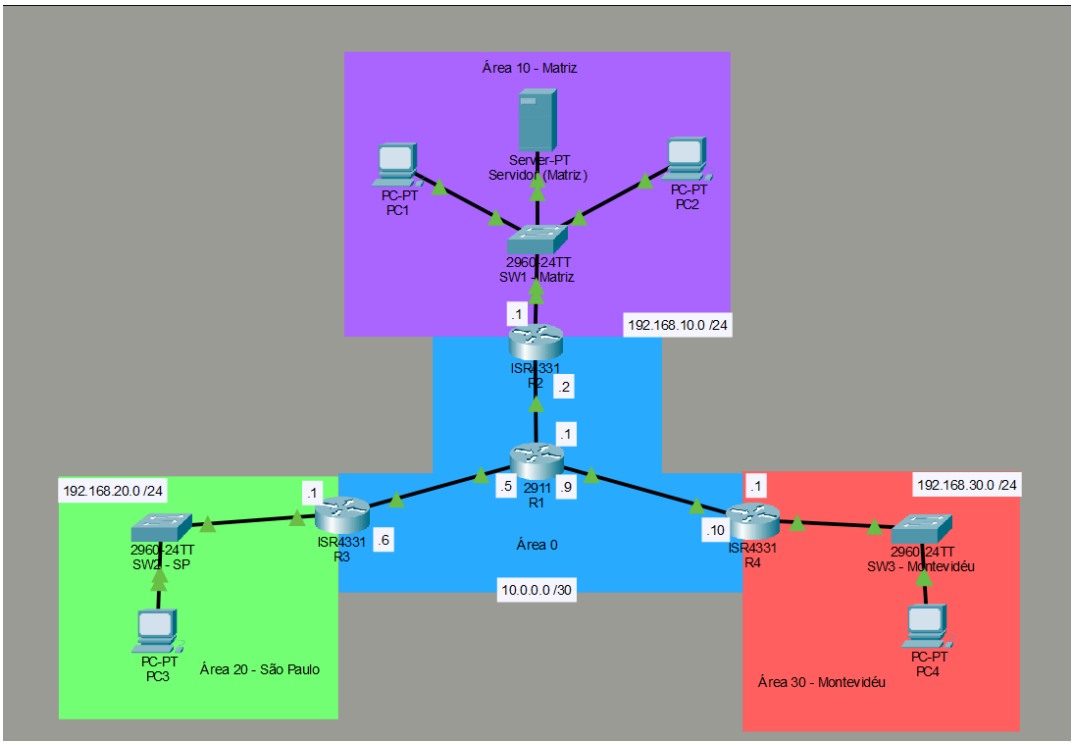
Projeto 4 — OSPF Multiárea: Matriz e Filiais

Criada em | October 13, 2026 3:19 PM

Topologia:



Todas as interfaces ativas:



Definindo o ip de cada roteador:

```
R2(config)#interface gigabitEthernet 0/0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
```

```
R3(config)#interface g0/0/0
R3(config-if)#ip address 10.0.0.6 255.255.255.252
R3(config-if)#exit
```

```
R4(config)#interface g0/0/0
R4(config-if)#ip a
R4(config-if)#ip add
R4(config-if)#ip address 10.0.0.10 255.255.255.252
R4(config-if)#
```

R1 Central:

```
Router(config-if)#exit
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#exit
Router(config)#hostname R1
R1(config)#interface g0/1
R1(config-if)#ip address 10.0.0.5 255.255.255.252
R1(config-if)#exit
R1(config)#interface g0/2
R1(config-if)#ip address 10.0.0.9 255.255.255.252
R1(config-if)#
```

Conexão estabelecida entre os roteadores (comprovado com ping)

```
R1#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#ping 10.0.0.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.6, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R1#ping 10.0.0.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R1#
```

ativo o ospf no roteador central, e com o comando network indico as interfaces e informo a área delas, para participarem do ospf:

```
R1(config)#router ospf 10
R1(config-router)#network 10.0.0.1 0.0.0.0 area 0
R1(config-router)#network 10.0.0.5 0.0.0.0 area 0
R1(config-router)#network 10.0.0.9 0.0.0.0 area 0
```

validando com o comando show ip ospf interface brief que as interfaces corretas estão ativadas para ospf:

```
R1(config)#do show ip ospf interface brief
Interface     PID   Area                    IP Address/Mask          Cost  State  Nbrs F/C
Gig0/0         10   0                     10.0.0.1/255.255.255.252   1       DR  0/0
Gig0/1         10   0                     10.0.0.5/255.255.255.252   1       DR  0/0
Gig0/2         10   0                     10.0.0.9/255.255.255.252   1       DR  0/0
```

repeti o processo em todos os outros roteadores nas interfaces necessarias.

alterei o id de todos os roteadores:

```
R1(config-router)#router-id 1.1.1.1
R1(config-router)#Reload or use "clear ip ospf process" command, for this to take effect
```

vizinhança entre ospf estabelecida, comprovando com o comando show i´p ospf neighbor:

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:32    10.0.0.6        GigabitEthernet0/1
4.4.4.4           1   FULL/BDR        00:00:36    10.0.0.10       GigabitEthernet0/2
2.2.2.2           1   FULL/DR         00:00:32    10.0.0.2        GigabitEthernet0/0
R1#
```

ao usar o comando para ver a interface detalhada, percebo que as conexões entre os roteadores estão no modo broadcast

quando usei o comando show ip route ospf, para ver se as areas 10, 20 e 30, foram aprendidas pelo ospf me deparei com uma linha em branco, sinal que nenhuma rota tinha sido aprendida, quando usei o comando show ip route vejo apenas as rotas diretamente conectadas que são as da áreas wans:

```
R1#show ip route ospf
R1#
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
C       10.0.0.4/30 is directly connected, GigabitEthernet0/1
L       10.0.0.5/32 is directly connected, GigabitEthernet0/1
C       10.0.0.8/30 is directly connected, GigabitEthernet0/2
L       10.0.0.9/32 is directly connected, GigabitEthernet0/2
```

cheguei a conclusão que as únicas redes que eu anunciei no ospf foram as redes wan, e que as redes lan´s devem ser anunciadas também. Após saber disso faço as configurações necessárias.

```
R2(config)#router ospf 10
R2(config-router)#network 192.168.10.1 0.0.0.0 area 10
R2(config-router)#
```

```
R3(config)#router ospf 10
R3(config-router)#network 192.168.20.1 0.0.0.0 area 20
R3(config-router)#
```

```
R4(config)#router ospf 10
R4(config-router)#network 192.168.30.1 0.0.0.0 area 30
```

A saída dos comandos agora monstram as rotas aprendidas pelo ospf

```
R1>
R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
C       10.0.0.4/30 is directly connected, GigabitEthernet0/1
L       10.0.0.5/32 is directly connected, GigabitEthernet0/1
C       10.0.0.8/30 is directly connected, GigabitEthernet0/2
L       10.0.0.9/32 is directly connected, GigabitEthernet0/2
O IA 192.168.10.0/24 [110/2] via 10.0.0.2, 00:49:36, GigabitEthernet0/0
O IA 192.168.20.0/24 [110/2] via 10.0.0.6, 00:05:35, GigabitEthernet0/1
O IA 192.168.30.0/24 [110/2] via 10.0.0.10, 00:05:35, GigabitEthernet0/2

R1#show ip route ospf
O IA 192.168.10.0 [110/2] via 10.0.0.2, 00:49:43, GigabitEthernet0/0
O IA 192.168.20.0 [110/2] via 10.0.0.6, 00:05:42, GigabitEthernet0/1
O IA 192.168.30.0 [110/2] via 10.0.0.10, 00:05:42, GigabitEthernet0/2
```

configurei estaticamente o pc de todas as áreas e realizei o teste de ping:

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Rede OSPF funcionando.

Como boa prática eu coloca as interfaces vinculadas as lans em modo passivo para o roteador mão mandar mensagens hello para a lan e congestionar o tráfego e gerar consumo de cpu que não precisava, já que não há roteadores ospf nas lans:

```
R2>enable
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 10
R2(config-router)#passive-int
R2(config-router)#passive-interface g
R2(config-router)#passive-interface gigabitEthernet0/0/1
R2(config-router)#end
```

```
R3>enable
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 10
R3(config-router)#passive-interface g
R3(config-router)#passive-interface gigabitEthernet0/0/1
R3(config-router)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#
```

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router ospf 10
R4(config-router)#passive=interface g
R4(config-router)#passive=interface gi
R4(config-router)#passive=interface gigabitethernet0/0/1
                                    ^
```

como cada interface conecta apenas a outro roteador, decido alterar o tipo da rede para poin-to-point para otimizar a rede, desativando o processo de eleição e fazendo com que a rede fique mais veloz para convergir (se recuperar de quedas) e consome menos memória dos equipamentos.

```
R1(config)#interface g0/0
R1(config-if)#ip ospf net
R1(config-if)#ip ospf network po
R1(config-if)#ip ospf network point-to-point
R1(config-if)#
05:01:14: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
e
05:01:23: %OSPF-5-ADJCHG: Process 10, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
xit
R1(config)#interface g0/1
R1(config-if)#ip ospf network point-to-point
R1(config-if)#
05:01:29: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/1 from FULL to DOWN, Neighbor Down: Interface down or detached

05:01:29: %OSPF-5-ADJCHG: Process 10, Nbr 3.3.3.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
exit
R1(config)#interface g0/2
R1(config-if)#ip ospf network point-to-point
R1(config-if)#
05:01:41: %OSPF-5-ADJCHG: Process 10, Nbr 4.4.4.4 on GigabitEthernet0/2 from FULL to DOWN, Neighbor Down: Interface down or detached

05:01:41: %OSPF-5-ADJCHG: Process 10, Nbr 4.4.4.4 on GigabitEthernet0/2 from LOADING to FULL, Loading Done
exit
```

valído que há adjacencia entre os vizinhos sem o processo de eleição com o comando show ip ospf neighbor:

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:35    10.0.0.6        GigabitEthernet0/1
4.4.4.4           0   FULL/  -        00:00:36    10.0.0.10       GigabitEthernet0/2
2.2.2.2           0   FULL/  -        00:00:36    10.0.0.2        GigabitEthernet0/0
R1#
```

com o comando show ip ospf interface comprovo que todas as conexões estão no modo point-to-point:

```
R1#show ip ospf interface

GigabitEthernet0/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 10, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 1
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 10, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 1
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:02
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/2 is up, line protocol is up
  Internet address is 10.0.0.9/30, Area 0
  Process ID 10, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 1
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 4.4.4.4
  Suppress hello for 0 neighbor(s)
R1#
```

valído que tudo está funcionando com um ping:

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=19ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125
Reply from 192.168.20.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 19ms, Average = 4ms

C:\>ping 192.168.30.2

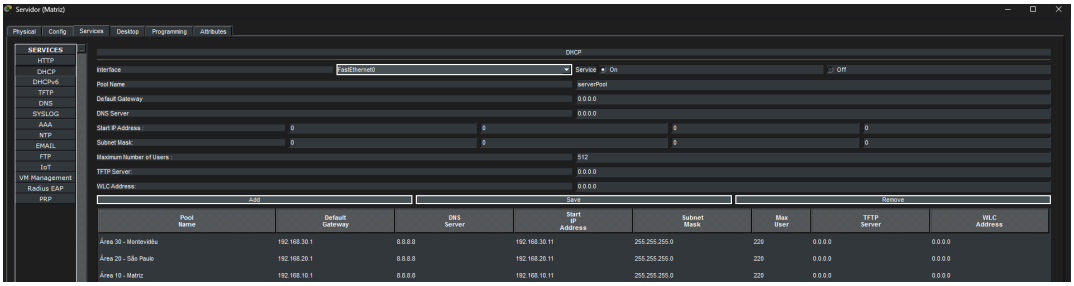
Pinging 192.168.30.2 with 32 bytes of data:

Reply from 192.168.30.2: bytes=32 time<1ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125
Reply from 192.168.30.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Após a rede estar 100% funcional, configuro o servidor dhcp:



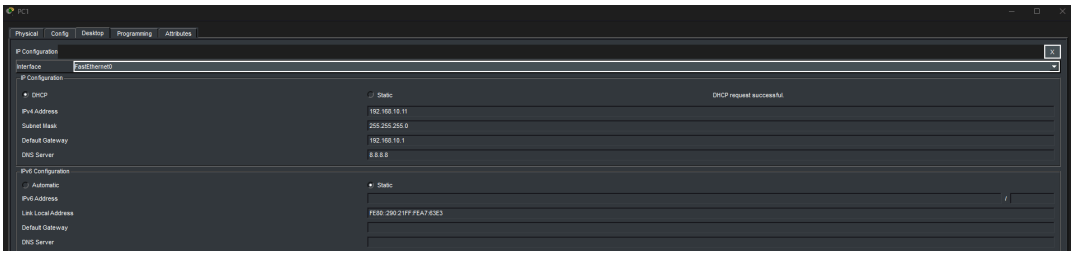
informo os outros roteadores que o servidor dhcp está na área 10:

```
R3(config)#interface gigabitEthernet 0/0/1
R3(config-if)#ip helper-adress 192.168.10.10
                               ^
% Invalid input detected at '^' marker.

R3(config-if)#ip helper-address 192.168.10.10
R3(config-if)#
```

```
R4(config)#interface gigabitEthernet 0/0/1
R4(config-if)#ip helper-address 192.168.10.10
R4(config-if)#
```

Faço a validação do servidor em todas as áreas:







5 horas de projeto desde a criação da topologia, plano de endereçamento até a validação de todas as conexões.